

AVANISH GADHIKAR

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EDUCATION

CARNEGIE MELLON UNIVERSITY, TEPPER SCHOOL OF BUSINESS

Pittsburgh, PA

Master of Science in Computational Finance – MSCF

GRE Quant: 170/170

12/27

- Upcoming courses in financial data science, options, investments, fixed income, stochastic calculus

INDIAN INSTITUTE OF TECHNOLOGY, GUWAHATI

Assam, India

Bachelor of Technology in Mathematics and Computing

CGPA: 9.01/10.0

06/26

- **Minor** in Data Science and Artificial Intelligence (DS & AI)
- Key courses included linear algebra, calculus, option pricing, machine learning, probability & statistics
- **TA** for a course 'Financial Valuation and Portfolio Analytics'; part of a study group on 'DeFi and FinTech'
- **Programming skills:** Python, C, C++ (proficient), MATLAB (intermediate)

RESEARCH/PROJECTS

MODELING OF MICROSTRUCTURES OF ELECTRONIC MARKETS

08/25 – 04/26

- **Probabilistic Modeling:** Simulated Limit Order Book dynamics through Poisson processes to model independent order flows; calibrated model parameters with LOBSTER tick data to replicate empirical market depth profiles.
- **Statistical Estimation:** Modeled self-exciting market events via a 6-dimensional Hawkes process; used Maximum Likelihood Estimation to obtain parameters for mid-price simulation, inter-arrival times.
- **Generative Modeling:** Implemented a Wasserstein Generative Adversarial Network (W-GAN) to generate synthetic order book states by learning conditional state distributions; evaluated the network on MSFT tick data against marginal densities of real market states.

MONTE CARLO SIMULATIONS

08/24 – 11/24

- **Mathematical Simulations:** Built simulations for random sampling and estimation using Inverse Transform, Acceptance-Rejection, and Box-Muller methods; achieved variance reduction through Antithetic Variables, Control Variates, and Stratification.
- **Optimizations & Pricing:** Modeled Geometric Brownian Motion and Jump Diffusion processes to price European, American and Asian options; utilizing Sobol, and Halton sequences for faster convergence.

xv6 OPERATING SYSTEM

08/24 – 11/24

- **Process Scheduling:** Extended the xv6 kernel to include three custom schedulers (Lottery, SJF, and Hybrid); implemented lazy allocation with an LRU-based swapping to optimize kernel memory.
- **File Storage:** Evaluated system Input/Output performance across two file systems (ZFS and EXT4), profiling write throughput, deduplication efficiency, and CPU overhead via system benchmarking tools.

EXPERIENCE

SALESFORCE

Bengaluru, India

Associate Member of Technical Staff, Intern

05/25 - 08/25

- **Agentic Systems:** Engineered Python AIOps package comprising 7 diagnostic tools within the Warden framework, automating Splunk and Argus queries to reduce on-call Time To Diagnose from hours to 2–4 minutes.
- **API Integration:** Architected custom Slack-integrated applications, to proactively track channel metrics, alert relevant personnel, and accelerate the resolution of outbound access and 403, 503 type errors.
- **Production Reliability:** Collaborated across infrastructure teams to ensure compatibility with workflows and high-availability environments; gaining experience in secure cloud architectures and live operations.

ADDITIONAL INFORMATION

- **Interests:** Football (Man Utd), Running (10k PB 55mins), F1 (Super Max), Music, Poker
- **Leadership:** Technical Secretary of Barak Hostel, IITG (Led a contingent of 100+ people to victory in an inter-hostel technical competition); Captained the high-school football team.
- **Languages:** Hindi, Marathi (Native)